# Subham Anand

Final Year Undergraduate
Department of Statistics and Data Science
Indian Institute of Technology Kanpur

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| Examination | Degree/Certificate | Institute | Year | CPI/% |
|---|---|---|---|---|
| Graduation | B.S.-SDS, Minor-CSE | Indian Institute of Technology Kanpur | 2022-2026 | **9.2**/10 |
| Intermediate/+2 | CBSE(XII) | Vikas Vidyalaya,Begusarai | 2022 | **97.2%** |
| Matriculation | CBSE(X) | DAV Public School,Itwa,Begusarai | 2020 | **97.0%** |

## Scholastic Achievements

- Awarded **Academic Excellence Award twice** for exceptional performance in academics for 2022 and 2022-23, IIT Kanpur
- Secured **All India Rank 870**, **JEE Advanced 2022** conducted by IIT Bombay, among 1,60,000 shortlisted candidates
- Secured **All India Rank 531** , **JEE Mains 2022** among the 15 Lakh candidates, conducted by National Testing Agency(NTA)
- Awarded **Inspire** Scholarship, in the years 2022,2023,2024,2025 for maintaining good academic performance consistently
- Awarded **Full Scholarship**, 6th Annual Nepal **AI School**, Completed 10 days of intensive lectures and labs led by Researchers/Professors from **Oxford**, **NUS** etc.

## Work Experience

**Samsung Research Institute Bangalore** — *May'25 - Jul'25*

*Developer Intern*

- Utilized **LoRA via Unsloth** for initial fine-tuning; transitioned to alternative frameworks for newer model compatibility.
- Designed and deployed a **scalable multi-GPU training pipeline for full fine tuning** on AWS using p5 instances to fine-tune large-scale LLMs, including Qwen-3-14B and Qwen-3-32B which are ≈30GB and ≈70GB in full precision respectively.
- Integrated **DeepSpeed ZeRO-2/3** to optimize memory efficiency for these large models and mitigate offloading overhead.
- Built a custom **Docker-based training environment** (Ubuntu, CUDA) for **AWS SageMaker** training jobs, enabling a reproducible and reusable fine-tuning workflow.
- **Quantized a 70GB fine-tuned Qwen-3-32B model down to 20GB** using vLLM's LLMCompressor (AWQ), significantly reducing memory footprint and making large-scale inference practical.
- Optimized training infrastructure by resolving **CUDA OOM error** and storage bottlenecks like **Docker-NVMe store mismatches** while leveraging **AWS Spot Instances** for cost-effective, scalable performance.

## Projects

**Open Source Work - Pre Release**

*Core Contributor* — Feb. 2026 - Present

- **High-Density Knowledge Extraction:** Architected a custom **Open Information Extraction (OpenIE)** pipeline for dense **Knowledge Graph (KG)** generation from unstructured text, optimizing entity/relation recall beyond the constraints of SimpleKGPipeline and LLMGraphBuilder.
- **LLMs Infrastructure:** Evaluated and benchmarked extraction performance across diverse model architectures (including **Qwen 3.7 Max, DeepSeek V4 Pro,** and **Kimi**); assessed cost vs output quality for these API's routed through **OpenRouter** to find the best fit for the purpose.
- **Eval Pipeline:** utilized **Label Studio** on an initial subset to design comprehensive evaluation rubrics, establishing inter-annotator agreement baselines via **Cohen's Kappa** to benchmark the final **LLM-as-a-judge** scoring stage

## Projects

**Neural Machine Translation for English-to-Indic** — *2025*

*Course Proj, Mentor: Prof. Ashutosh Modi, Dept. of CSE, IIT Kanpur*

- Built an **English-to-Hindi/Bengali NMT** using a **3-layer Transformer** ($d_{model}$ = 512, 8 heads), optimizing training with **Cross-Entropy Loss** on 149k pairs preprocessed via **spaCy/IndicNLP**.
- Mitigated OOV bottlenecks using **cross-attention weights** and **POS tagging** for transliteration, achieving a final **BLEU score of 0.166** and **chrF of 0.430** on unseen test sets.

**Natural Language Processing & Neural Architectures** — *2025*

*Course Assign., Mentor: Prof. Ashutosh Modi, Dept. of CSE, IIT Kanpur*

- Executed foundational NLP methodologies: verified **Zipf's Law** across multilingual corpora, engineered **Regex** and **spaCy** pipelines for **POS/NER tagging**, and implemented **BPE**, **WordPiece**, and **Unigram** tokenizers from scratch.
- Developed **Word2Vec (Skip-gram/CBOW)** using **Negative Sampling** alongside **EM algorithms** for document clustering; independently pretrained and fine-tuned a **causal character-level PyTorch Transformer** for spell correction and sentiment classification.

**Bayesian Effect Fusion for Categorical Predictors** — *2025*

*Course Project: MTH442, Prof. Arnab Hazra, IIT Kanpur*

- Implemented Bayesian effect fusion in R and benchmarked **glmnet**, **monomvn**, and **grpreg** to evaluate performance.
- Executed **MCMC via Gibbs sampling**; used **Binder's loss** to identify optimal fusions of redundant categorical levels.
- Applied model to **Kaggle Airline data**, reducing parameter space from 55 levels to 7 coefficients while maintaining accuracy and also attained the lowest **MSPE** among all methods

**Clinical Trial Design Using Cancer Data** — *2025*

*Mentor: Prof. Dootika Vats, Dept. of Statistics, IIT Kanpur — In collaboration with Johnson & Johnson*

- Performed large-scale **survival analysis** and **gene mutation profiling** on 25,000 multi-cancer patient records.
- Designed and simulated a **lung cancer clinical trial**, optimizing sample size and statistical power with **hazard modeling**.
- Applied **Bayesian interim analysis** to treatment response, establishing early stopping criteria using **posterior probability thresholds**.

## Positions Of Responsibility

**Secretary, Programming Club — SnT Council** — *2023 - 2024*

- Showcased **BeEF** and phishing frameworks, co-organized **Linux Install Fest**, and designed campus engagement posters. and developed**InfoSec self-assessment tasks**

## Technical Skills

- **Languages & Frameworks:** Python, R, C++, PyTorch, spaCy, Hugging Face, Unsloth, Scikit-learn, Git, Canva, Word2Vec, neo4j, Cypher, SQL,.
- **AI/NLP :** Transformers, GNNs, Word Embeddings, BPE/WordPiece Tokenisation, Visual Language Models.
- **Cloud & MLOps:** AWS (SageMaker, p5 & g5 instances , Spot), DeepSpeed (ZeRO-2/3), Docker.
- **Stats & Analytics:** Bayesian Modeling, MCMC, EM Algorithm, Zipfian Analysis, Hypothesis Testing.

## Relevant Courses

(* Ongoing)

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|---|---|---|---|
| Time Series Analysis | Statistical Computing | Randomized Algorithms | Principles of DBMS |
| Intro to Machine Learning | Data Structures & Algorithms | Markov Chain Monte Carlo | Introduction to Probability Theory |
| Natural Language Processing | Stochastic Processes | Principles of Program Lang. | Matrix Algebra and Linear Est. |